Test-procedure choice for the biomarker-treatment interaction in aggregated N-of-1 trials: dichotomization, classical RM-ANOVA, mixed-effects, and generalized estimating equations

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Contents

1 Abstract

Background. Inference about the biomarker-treatment interaction in aggregated N-of-1 trials depends on the analyst's test procedure: classical repeated-measures ANOVA (RM-ANOVA), the linear mixed-effects model (LME), or generalized estimating equations (GEE). A common but rarely examined feature of the classical analysis is that it requires the continuous biomarker to be dichotomized. We quantify the consequences of test-procedure choice for the interaction, separating the cost of dichotomization from the cost of the test procedure proper.

Methods. We derive the closed-form strict RM-ANOVA F-statistic for the biomarker-stratified treatment-by-time interaction in a balanced split-plot design with categorical predictors and the sphericity assumption, and relate it to the Wald t-test for the corresponding fixed-effects coefficient in a linear mixed model with continuous-time AR(1) residual correlation. We then conduct a Monte Carlo simulation study, designed and reported under the[1] ADEMP framework, that crosses five test procedures (strict RM-ANOVA on the dichotomized biomarker; a continuous-biomarker within-subject contrast; the standard pmsimstats linear-mixed analysis with `corCAR1`; and GEE with naive and with Mancl-DeRouen small-sample sandwich variance) against the biomarker-moderation coefficient $\beta_{bm} \in \{0, 0.45\}$ and sample size $N \in \{25, 35, 70, 100\}$, a 40-cell factorial at the prazosin-PTSD reference design with 1000 replicates per power cell and 2000 per Type I cell. The continuous-biomarker contrast differs from strict RM-ANOVA only in retaining the continuous biomarker, isolating the cost of dichotomization. The cycle-by-period design grid is specified here but its empirical sweep is deferred to a companion paper.

Results. The closed-form strict RM-ANOVA test agrees with the linear-mixed Wald test under sphericity and balanced design. In the simulation, most of the power gap between strict RM-ANOVA and the mixed model is the cost of dichotomization rather than of the test procedure: at $N = 25$, replacing the dichotomized biomarker with the continuous one raises power by 0.180, while the further step to the full mixed model adds only 0.027, a pattern stable across N (dichotomization 0.08–0.21, test procedure at most 0.04). Among the four continuous-biomarker procedures, the linear-mixed analysis attains the highest power within nominal Type I error; GEE with the naive sandwich inflates Type I error to between 1.5× and 1.9× nominal across all four sample sizes, which the Mancl-DeRouen correction removes at a moderate (about 10-percentage-point) power cost while restoring 93–95% interval coverage. The inline Mancl-DeRouen standard error agrees with the `geesmv` reference to within 2.5%.

Conclusions. The analyst's most consequential decision is to keep the biomarker continuous rather than dichotomize it; among continuous-biomarker procedures the linear-mixed analysis is the preferred default and a dichotomization-free within-subject contrast is a close, transparent alternative. Naive GEE should not be used at the sample sizes examined without

a small-sample correction. This paper resolves the test-procedure axis at a fixed reference design; the joint cycle-by-period design grid is specified here and deferred to a companion paper.

2 Introduction

2.1 A claim about how design and test choices interact

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The present paper develops the methodological case for treating the analyst's two principal choices in an aggregated N-of-1 trial jointly rather than separately. The two choices are the test procedure for the biomarker-treatment interaction (strict repeated-measures ANOVA, the linear mixed-effects model with continuous-time AR(1) residual correlation, the mixed-effects model for repeated measures, generalized estimating equations) and the trial-design parameters that govern within-subject treatment cycling (number of treatment cycles, on-drug and off-drug period lengths, on/off symmetry, optional run-in and run-out phases). The validity and power of any given test depend on what the design assumes about carryover, autocorrelation, and within-subject variance structure; conversely, the optimal design parameters depend on which test the analyst plans to use. Treating the two as independent optimizations, as the published methodological literature typically does, produces locally optimal but jointly suboptimal combinations, and we shall argue that this gap is consequential at the sample sizes characteristic of precision-medicine pilot studies.

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The case we develop is two-part. We begin by establishing that a joint design-by-test sensitivity study has not, to the best of our knowledge, been performed for the aggregated N-of-1 design class, and that the absence of one limits the field's ability to make defensible joint recommendations. We then demonstrate, through a focused Monte Carlo experiment, that the linear mixed-effects test with continuous-time AR(1) residual correlation, used in conjunction with the Hendrickson hybrid open-label-plus-blinded-discontinuation design[2], dominates the strict classical RM-ANOVA test in essentially every parameter cell where within-participant residuals depart from compound symmetry. That is, strict RM-ANOVA remains useful as a stress test — the most stringent baseline that compound-symmetry machinery can deliver — but not as a recommendation in its own right.

2.2 Why the question matters for chronic-condition pharmacology

In this section we shall set out three reasons why the joint treatment of test procedure and design parameters is warranted specifically for the aggregated N-of-1 design class.

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We begin with the physiological motivation. Within-participant residuals in chronic-condition trials are autocorrelated not as an analytic convenience but because most chronic-condition response trajectories exhibit serial correlation on a timescale set by drug pharmacokinetics, condition-specific natural-history drift, and measurement reactivity (Hawthorne, structured-visit, and diary-completion effects). Continuous-time AR(1) residual correlation is, in our view, the appropriate structural prior for this physiology, but it does not satisfy the sphericity assumption required by strict RM-ANOVA.[3] show through simulation that uncorrected RM-ANOVA inflates Type I error under sphericity violations; they also report, importantly, that the Greenhouse-Geisser-corrected RM-ANOVA and the multilevel linear model perform comparably once the correction is applied, and that the multilevel model is not uniformly safer (it can be the more liberal of the two in some conditions). The degrees-of-freedom corrections[4,5] recover nominal levels only conditionally.[6] extend this evidence with finer epsilon resolution, with the conventional rule favoring Greenhouse-Geisser when $\epsilon < 0.75$. The advantage we claim for the mixed-effects model with explicit `corCAR1` residual structure is thus not that it is uniformly more powerful than a well-corrected RM-ANOVA, but that it models the AR(1) structure directly, accommodates the continuous biomarker and unequal time spacing, and so avoids both the dichotomization and the degrees-of-freedom penalty that the classical route incurs.

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The biomarker-treatment interaction also merits separate attention because it is harder to detect than the average treatment effect (ATE). The PATH framework[7,8] makes the methodological distinction precise: effect-modeling (the predictive heterogeneity-of-treatment-effects, or HTE, question) requires more design-aware machinery than risk-modeling (the ATE question). The within-participant contrasts that contribute to the interaction estimator are sensitive to the trial-design parameters in ways the ATE analysis is not, and a test procedure that underpowers the interaction term will systematically miss predictive-biomarker signal even at sample sizes that appear adequate for the ATE test.

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Finally, cycle count and period length jointly determine the within-subject contrast structure in ways that cannot be optimized independently.[9] and[10] establish, in the broader cross-over literature, that adding within-subject treatment cycles contributes information about the patient-by-treatment interaction variance that no cross-sectional comparator can supply. The N-of-1 specialization of this principle is that more cycles increase power for the biomarker-treatment interaction, but with diminishing returns: cycles beyond the fourth or fifth contribute marginal information at the cost of trial duration. Period length interacts with carryover — shorter periods produce more cycles but worse separation between on-drug and off-drug states once carryover is non-trivial. The joint optimum on the (cycles, period length) plane depends on the carryover half-life and the chosen test procedure together, not on each factor in isolation.

2.3 What the published N-of-1 methodology literature does and does not do

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We begin with the state of the analytic-model literature. The systematic review of[11] documents that 83.8% of the published N-of-1 studies they catalogue ignore autocorrelation entirely;[12] and[13] reach similar conclusions in neurology and behavioral-medicine subdomains. CONSORT extension reporting standards[14,15] codify reporting quality but do not address the analytic-model decision in any detail, and[16] identifies the analytic-method gap as the highest-priority methodological problem in the field. It appears, then, that the majority of N-of-1 analyses are conducted under a misspecified residual covariance assumption, and that the field has not yet converged on remedies.

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Test-procedure comparisons for longitudinal data are not new. The generic contrast of RM-ANOVA against GEE and mixed-effects models is well established:[17], for instance, report through simulation that RM-ANOVA is appreciably less powerful than GEE and mixed models for continuous longitudinal endpoints, the same ordering we recover. What that literature does not do is treat the biomarker-treatment interaction estimand, the aggregated N-of-1 / cross-over design, or the continuous-time AR(1) structure together, nor does it isolate the dichotomization cost that the classical route imposes specifically when the moderator is continuous. The MMRM consensus established by Mallinckrodt and colleagues[18–21] applies to parallel-group longitudinal trials with monotone missingness, not to aggregated N-of-1 designs. The GEE methodology of[22] with sandwich variance estimation has been adapted to small-cluster settings via bias-corrected estimators[23,24], but the small-sample concerns are acute at trial-relevant N-of-1 cluster counts:[25] report that in stepped-wedge cluster trials with fewer than six independent units only 6.6% of analyses use appropriate small-sample corrections, illustrating the gap between method availability and practice. None of these papers address the biomarker-interaction estimand in the cross-over or N-of-1 setting directly.

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Design-parameter optimization has been studied, but separately from test-procedure choice.[26] develop power calculations for $(AB)^k$ designs that match the N-of-1 within-subject contrast structure;[27] document strong autocorrelation effects on permutation-test power; and[10] demonstrates how replicate cross-over designs allow direct estimation of patient-by-treatment interaction variance, the conceptual ancestor of N-of-1 aggregation. The Hendrickson et al. framework[2] is, to the best of our knowledge, the only paper that compares full design families head-to-head for the biomarker-interaction power question, but it holds period length fixed and uses a single test procedure.[28] compare aggregated N-of-1 trials against parallel and cross-over designs by simulation and find the aggregated design more efficient, but analyze a single mixed model and do not vary the test procedure.[29] and[30] synthesize the broader N-of-1 design literature without addressing the joint design-by-test sensitivity question.

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The gap, then, is this. Across the recent systematic reviews[11,12,31] and the methodological literature surveyed above, no published study crosses test procedure (strict RM-ANOVA, MMRM, GEE, mixed-effects with `corCAR1`) against a design-parameter grid (cycles, period lengths, asymmetry, run-in, run-out) under simulation for the aggregated N-of-1 design class. The present paper endeavors to address that gap, at least on the test-procedure axis.

2.4 Run-in design and the placebo-selection controversy

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A separate methodological controversy that the joint design-by-test framing intersects concerns the use of placebo run-in phases to exclude placebo responders before randomization.[32] show formally that response-based exclusion during run-in introduces a selection bias that damages external validity and may affect internal validity as well. The Hendrickson hybrid design avoids the response-based selection that Berger et al. flag by using an open-label stabilization phase rather than a placebo run-in; the design choice is therefore conservative on the run-in question, though it forgoes the power that aggressive response-based selection would otherwise supply. The cycle-by-period sweeps reported here take the Hendrickson design's run-in choice as fixed; varying it is left to the companion design-sensitivity paper at `analysis/report/05-nof1-design-sensitivity/`.

2.5 What this paper contributes

We make four contributions, each addressed in a section below.

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First, we derive the closed-form strict RM-ANOVA F-statistic for the biomarker-stratified treatment-by-time interaction in a balanced split-plot design with categorical predictors and the sphericity assumption (§2). We relate the strict-RM-ANOVA test to the Wald t-test on the biomarker-treatment fixed-effects coefficient in a linear mixed model with continuous-time AR(1) residual correlation, identifying the conditions on residual covariance under which the two tests agree and the regimes in which they diverge. The derivation extends standard cross-over methodology[9,33] to the biomarker-stratified case and clarifies the connection to the contemporary sphericity-violation literature[3,6].

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Second, we specify a cycle-by-period design grid (§3) that varies the number of treatment cycles, on-drug and off-drug period lengths, and on/off symmetry around the Hendrickson hybrid reference. The grid is calibrated to the prazosin/PTSD reference application that motivates the pmsimstats program.

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Third, we conduct a Monte Carlo simulation study (§4–§5) that crosses five test procedures (strict RM-ANOVA on the dichotomized biomarker; a continuous-biomarker within-subject contrast; the standard pmsimstats linear-mixed analysis with `corCAR1`; and GEE with naive and with Mancl-DeRouen[34] small-sample sandwich variance) against the biomarker-moderation coefficient and sample size at the fixed reference design. The continuous-biomarker contrast is the dichotomization-free analogue of the strict RM-ANOVA test and lets us separate the cost of dichotomizing the biomarker from the cost of the test procedure proper. The simulation runs on the pmsimstats package data-generating process under the mean-moderation architecture via `generateData()` and `lme_analysis()`, holding fixed the response-curve and covariance-structure axes addressed in `analysis/report/07-gompertz-evaluation/` and `analysis/report/01-dgp-mean-moderation-vs-mvn/`. Results are reported in the[1] ADEMP framework adopted as the project-wide simulation-reporting standard.

Notation for the moderation parameter. Because the data-generating process here is the mean-moderation architecture throughout, the biomarker-moderation parameter is written β_{bm}, the dimensionless multiplier scaling the treatment effect by the standardized biomarker. The companion architecture manuscript reserves c_{bm} for the Architecture-B parameter, which is a correlation rather than a regression multiplier. The two are calibrated to a common numeric scale (the Architecture-A shift is $\beta_{bm}\, b_i\, \sigma_{BR}$ on the standardized biomarker b_i), so the reference value 0.45 denotes a matched moderation strength under either architecture, but the symbols are not interchangeable.

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Fourth, we synthesize a design-and-test recommendation table (§6) for typical chronic-condition N-of-1 trial scenarios. The recommendation is presented as a table of preferred (cycle count, period length, test procedure) combinations indexed by the carryover half-life and the target effect size, rather than as a single 'best' design.

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The remainder of the paper is organized as follows. Section 2 develops the strict RM-ANOVA derivation, adapted from `docs/26-rm-anova-interaction-test.md`. Section 3 specifies the cycle-by-period design grid, adapted from `docs/27-cycle-period-design-sweep-p` Section 4 lays out the simulation design. Section 5 reports the results. Section 6 synthesizes the joint design-and-test recommendation. Section 7 discusses the implications for the methodology of N-of-1 biomarker-validation trials and for the operating characteristics of the pmsimstats simulation program.

3 Closed-form strict RM-ANOVA test of the biomarker-treatment interaction

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In this section we shall derive the closed-form strict RM-ANOVA F-statistic for the biomarker-treatment interaction in a balanced split-plot design with categorical predictors and the sphericity assumption, and relate it to the Wald t-statistic on the corresponding linear-mixed coefficient. The derivation specializes classical split-plot machinery[35] to the biomarker-stratified case in the simplest design that supports the test: two biomarker strata, two treatment phases, and T within-phase timepoints. The presentation is deliberately confined to strict RM-ANOVA. Modern extensions — the mixed-effects model for repeated measures[36], generalized estimating equations[22], and ANCOVA with random subject effects — appear only for context; the closed-form expression we report is the textbook compound-symmetric form. Three findings anchor the rest of the paper:

1. Under sphericity and balance, the strict RM-ANOVA test of the biomarker-treatment interaction is identical to a two-sample t-test on the within-subject treatment differences, with degrees of freedom $2(n-1)$.
2. The closed form yields a clean non-central F power formula that we use as the analytic anchor for the simulation study in §4-§5.
3. Five assumptions of the closed form are violated by the pmsimstats data-generating process (DGP; continuous biomarker, AR(1) within-subject covariance, exponential carryover, missing data, unequal time spacing). Each violation favors the linear-mixed

analysis over the strict RM-ANOVA test in operating characteristics, motivating the primary-and-sensitivity reporting strategy of §6.

3.1 Notation and design

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We begin with the simplest design that supports a biomarker-treatment interaction test in a within-subject setting. The reader will note that the design structure described here is more restrictive than the pmsimstats DGP; we return to that gap explicitly in §2.4.

Specifically, we consider the following structural elements:

- Two biomarker strata $i \in \{1, 2\}$, formed by dichotomizing the continuous biomarker at a pre-specified cut-point. Strict RM-ANOVA requires categorical predictors; the dichotomization is the price of admission to the framework.
- n subjects per stratum, balanced. Index $l \in \{1, \dots, n\}$.
- Two treatment phases $j \in \{1, 2\}$, active drug and placebo. In an N-of-1 design these are two phases through which every subject passes; in a parallel-group design they are between-subject and the design is not a split-plot.
- T within-phase timepoints, indexed $k \in \{1, \dots, T\}$. In the simplest case $T = 1$, the analyst pre-averages within-phase observations to a single mean per subject per phase.

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The biomarker stratum is between-subjects (each subject sits in exactly one stratum); the treatment phase and the timepoints are within-subjects (each subject sees both phases at every timepoint). This is the canonical split-plot repeated-measures design[35].

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We write Y_{ijlk} for the symptom outcome at biomarker stratum i, treatment phase j, subject l within stratum i, and timepoint k. The strict classical split-plot model is

$$\begin{aligned} Y_{ijlk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \pi_{l(i)} \\ + \tau_k + (\alpha\tau)_{ik} + (\beta\tau)_{jk} + (\alpha\beta\tau)_{ijk} + e_{ijlk}, \end{aligned} \tag{1}$$

with sum-to-zero constraints on each fixed effect, the random subject effect $\pi_{l(i)} \sim N(0, \sigma^2_\pi)$ nested in stratum, and within-subject error $e_{ijlk} \sim N(0, \sigma^2_e)$. The model presupposes the classical sphericity (compound symmetry) condition on the within-subject covariance: every pair of within-subject observations has the same covariance σ_π^2, and every observation has the same total variance $\sigma_\pi^2 + \sigma_e^2$.

3.2 Sums of squares and the interaction F-statistic

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We begin the derivation by partitioning the total sum of squares. The total sum of squares partitions into between-subjects and within-subjects components:

$$SS_{\text{tot}} = SS_{\text{between}} + SS_{\text{within}}. \tag{2}$$

The between-subjects component decomposes further into SS_A (biomarker stratum) and $SS_{S(A)}$ (subject within stratum):

$$SS_A = bnT \sum_{i=1}^{a} (\bar{Y}_{i\cdots} - \bar{Y}_{\cdots})^2, \quad SS_{S(A)} = bT \sum_{i=1}^{a} \sum_{l=1}^{n} (\bar{Y}_{i\cdot l\cdot} - \bar{Y}_{i\cdots})^2. \tag{3}$$

The within-subjects component decomposes into SS_B (treatment phase), SS_{AB} (the biomarker-treatment interaction, the primary target of the test), and a within-subjects error term $SS_{B\times S(A)}$:

$$SS_{AB} = nT \sum_{i,j} (\bar{Y}_{ij\cdot\cdot} - \bar{Y}_{i\cdots} - \bar{Y}_{\cdot j\cdot\cdot} + \bar{Y}_{\cdots})^2, \tag{4}$$

$$SS_{B\times S(A)} = T \sum_{i,j,l} (\bar{Y}_{ijl\cdot} - \bar{Y}_{i\cdot l\cdot} - \bar{Y}_{ij\cdot\cdot} + \bar{Y}_{i\cdots})^2. \tag{5}$$

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Corresponding degrees of freedom are $df_A = a-1$, $df_{S(A)} = a(n-1)$, $df_B = b-1$, $df_{AB} = (a-1)(b-1)$, and $df_{B\times S(A)} = a(n-1)(b-1)$, the last being the within-subjects error degrees of freedom appropriate for the treatment effect and its interaction with the between-subjects factor. The biomarker-treatment interaction is then tested by the F-ratio

$$F_{AB} = \frac{MS_{AB}}{MS_{B\times S(A)}} = \frac{SS_{AB}/df_{AB}}{SS_{B\times S(A)}/df_{B\times S(A)}}. \tag{6}$$

Under the null $H_0 : (\alpha\beta)_{ij} \equiv 0$ and the sphericity assumption, $F_{AB} \sim F(df_{AB}, df_{B\times S(A)})$.

3.2.1 The 2×2 split-plot case

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We now specialize to $a = b = 2$ (two biomarker strata, two phases), setting $T = 1$ for cleanest exposition; that is, the analyst pre-averages within-phase observations to a single value per subject per phase before computing the interaction. The sample interaction contrast is

$$\hat{L} = (\bar{Y}_{11\cdot} - \bar{Y}_{12\cdot}) - (\bar{Y}_{21\cdot} - \bar{Y}_{22\cdot}), \tag{7}$$

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That is, $\hat{L}$ is the difference of within-subject treatment differences across biomarker strata, and is the natural biomarker-treatment interaction statistic. The interaction parameter under the sum-to-zero ANOVA parameterization satisfies $\widehat{(\alpha\beta)}_{11} = \hat{L}/4$ (and the other three cells take values $\pm\hat{L}/4$), so the interaction sum of squares is $SS_{AB} = nT\,\hat{L}^2/4$, and with $df_{AB} = 1$ this equals the interaction mean square. Under sphericity and balance, $MS_{B\times S(A)} = \hat{\sigma}_e^2$. Substituting into equation (6) yields

$$F_{AB} = \frac{nT\,\hat{L}^2}{4\,\hat{\sigma}_e^2}, \qquad df_1 = 1, \quad df_2 = 2(n-1). \tag{8}$$

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Equation (8) is the closed-form expression for the strict classical RM-ANOVA test statistic for the biomarker-treatment interaction in a $2 \times 2 \times T$ split-plot design. We shall refer to it as the 'strict RM-ANOVA F' throughout the rest of the paper, to distinguish it from the mixed-effects and GEE Wald statistics derived from the same data.

3.2.2 Equivalence to a within-subject t-test

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We note that equation (8) is equivalent to a two-sample t-test on the within-subject differences. Define $\Delta_{il} = \bar{Y}_{i1l\cdot} - \bar{Y}_{i2l\cdot}$ as the within-subject treatment difference for subject l in biomarker stratum i, averaged over T within-phase timepoints. Then

$$\hat{L} = \bar{\Delta}_1 - \bar{\Delta}_2, \qquad \bar{\Delta}_i = \frac{1}{n}\sum_{l=1}^{n} \Delta_{il}. \tag{9}$$

Under sphericity, Δ_{il} are i.i.d. within stratum with $\mathrm{Var}(\Delta_{il}) = 2\sigma_e^2/T$, since the random subject effect cancels in the within-subject difference. The variance of $\hat{L}$ is therefore

$$\mathrm{Var}(\hat{L}) = \frac{2 \cdot 2\sigma_e^2}{nT} = \frac{4\sigma_e^2}{nT}, \tag{10}$$

and the t-statistic for the contrast is

$$t_{AB} = \frac{\hat{L}}{\sqrt{4\hat{\sigma}_e^2/(nT)}} = \sqrt{\frac{nT}{4}} \cdot \frac{\hat{L}}{\hat{\sigma}_e}, \tag{11}$$

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We see that $t_{AB}^2 = F_{AB}$, recovering equation (8). The strict-RM-ANOVA F-test for the biomarker-treatment interaction is therefore identical to a two-sample t-test on the within-subject treatment differences, with degrees of freedom $2(n-1)$. This identity is well-known in the split-plot literature[35] and is the analytical anchor for the statement that the strict-RM-ANOVA test of a biomarker-treatment interaction reduces to a simple two-sample comparison of treatment-effect estimates between biomarker strata.

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That is, under sphericity and balance, the biomarker-treatment interaction F-test asks simply whether the within-subject treatment effect differs between biomarker strata, and the answer is given by a two-sample t-test on those within-subject differences. This reduction has intuitive appeal and makes the test's assumptions unusually transparent: anything that biases or inflates the within-subject differences will distort the test in a predictable direction. We shall exploit this transparency when relating the strict-RM-ANOVA results to the mixed-effects results in §5.

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For a biomarker strata, b treatment phases, n subjects per stratum, and T within-phase timepoints, equation (6) generalizes directly with degrees of freedom $df_1 = (a-1)(b-1)$ and $df_2 = a(n-1)(b-1)$. Including the time factor τ_k explicitly, the test of whether the biomarker modifies the treatment effect across time is the three-way interaction $(\alpha\beta\tau)_{ijk}$; the corresponding F-statistic is $F_{AB\tau} = MS_{AB\tau}/MS_{B\tau\times S(A)}$ with $df_1 = (a-1)(b-1)(T-1)$ and $df_2 = a(n-1)(b-1)(T-1)$.

3.3 Relation to the linear-mixed Wald test

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We shall now relate the strict-RM-ANOVA F-statistic of equation (8) to the pmsimstats `nlme::lme` Wald t-statistic on the `bm:Dbc` coefficient. The two are asymptotically equivalent under three conditions: (i) the biomarker is dichotomized at the cut-point that defines the strata, (ii) the within-subject covariance is exactly compound-symmetric, and (iii) `Dbc` is binary, the E1 specification of the carryover-sensitivity manuscript rather than the matched-decay E2 specification. When all three conditions hold, the two tests reach the same coefficient and the same standard error up to small-sample correction factors. When any one fails, the mixed-model test extracts more information than the strict RM-ANOVA: the continuous biomarker contributes within-stratum variation, the AR(1) covariance is modeled directly, and the continuous exposure-decay predictor draws on the off-drug trajectory.

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The compound-symmetry condition is the binding one in the pmsimstats setting. Equation (8) presupposes sphericity: the within-subject covariance matrix is compound-symmetric. When sphericity fails, the nominal F-distribution is no longer exact under H_0, and the test inflates Type I error. Two diagnostics and two corrections are standard. Mauchly's test[37] is a likelihood-ratio test for compound symmetry against an unstructured alternative; it often rejects in moderate samples but is not informative about the direction of departure. Box's ϵ, with $\epsilon = 1$ at compound symmetry and $\epsilon \geq 1/(T-1)$ at maximum departure, is a scalar measure of departure from sphericity. The Greenhouse-Geisser correction[5] estimates $\hat{\epsilon}$ from the observed within-subject covariance and adjusts the degrees of freedom: $F_{AB} \sim F(\hat{\epsilon} \cdot df_1, \hat{\epsilon} \cdot df_2)$ under H_0. The Huynh-Feldt correction[4] is a less-conservative variant recommended when $\hat{\epsilon} > 0.75$. The contemporary sphericity-violation literature[3,6] documents substantial Type I error inflation when sphericity violations go uncorrected, with the conventional rule favoring Greenhouse-Geisser when $\hat{\epsilon} < 0.75$.

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The pmsimstats DGP imposes AR(1) within-factor correlation, which is not compound-symmetric. Under AR(1), Box's ϵ is strictly below 1, and treating the full multi-timepoint series as repeated within-subject levels under classical RM-ANOVA inflates Type I error unless a degrees-of-freedom correction is applied. The relevant magnitude is recorded in the project's own audit (`docs/06-ar1-residual-correlation.tex`): at $T = 8$ timepoints, fitting a mixed model with no residual-correlation structure (the naive analogue of treating AR(1) data as exchangeable) produces Type I error of 13 to 17 percent for the crossover (CO) and open-label (OL) designs, which switching to `nlme::lme` with `corCAR1` returns to nominal. We stress that this 13-to-17-percent figure is for the unstructured/naive fit, not for the strict RM-ANOVA arm evaluated in this paper: that arm pre-averages each phase to a single value (the 2×2 form of §2.2), so its within-subject covariance is 2×2, sphericity holds by construction, and its

Type I error is at nominal (§5). The sphericity penalty is therefore a property of the un-pre-averaged classical analysis; for the pre-averaged strict RM-ANOVA the operative cost is dichotomization, not sphericity. Either way the linear-mixed analysis with `corCAR1` avoids both costs, modeling the AR(1) structure directly rather than absorbing it via a degrees-of-freedom penalty.

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For the founding pmsimstats interaction question[2], the strict-RM-ANOVA test is therefore a conservative reduction of the available analysis rather than a competing alternative. Where a reviewer requests 'RM-ANOVA results', three responses are defensible: (a) point at the existing linear-mixed analysis as a continuous-time variant of the RM-ANOVA family, (b) report the strict-RM-ANOVA test from equation (8) as a sensitivity comparator under dichotomized biomarker and binary phase, or (c) report both with explicit power-loss tabulation. The simulation study in §4–§5 operationalizes option (c).

3.3.1 Power calculation under the closed form

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For the 2×2 case, equation (8) yields a closed-form power formula that we shall use as the analytic anchor for the simulation study in §4–§5. Under the alternative $H_1 : \hat{L} = \delta$, the statistic $F_{AB} = nT\,\hat{L}^2/(4\hat{\sigma}_e^2)$ follows a non-central F distribution with non-centrality parameter

$$\lambda = \frac{nT\,\delta^2}{4\sigma_e^2}, \tag{12}$$

and power at level α is

$$1 - \beta = \Pr\Big[\, F_{AB} > F_{1,2(n-1),1-\alpha} \,\Big|\, \lambda \,\Big], \tag{13}$$

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computable via standard non-central F functions (`stats::pf` in R with the `ncp` argument; `PROC POWER` in SAS). For sample-size calculation, equation (12) inverts to $n = 4\sigma_e^2\lambda/(T\delta^2)$, with λ chosen to deliver the target power at the design α. The formula assumes (i) sphericity, (ii) categorical biomarker at the dichotomized cut-point with equal stratum sizes, (iii) no carryover, (iv) no within-treatment relapse, and (v) i.i.d.
within-phase observations after within-subject differencing. All five assumptions are violated to varying degrees in the pmsimstats DGP, as we shall develop in the next subsection.

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We note that these five assumptions are not independent: violations of sphericity (i) and of the i.i.d. within-phase assumption (v) are, in the AR(1) setting, two faces of the same departure from compound symmetry. The analytic power formula should therefore be understood as an upper bound on the power available from the strict-RM-ANOVA test under the pmsimstats DGP, not a calibrated prediction.

3.3.2 Limitations of the strict classical formulation

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Unfortunately, the closed-form simplicity of equation (8) is bought at the price of five strong assumptions, each of which is empirically problematic in the pmsimstats setting. We present them in approximate order of severity.

1. *Categorical biomarker.* The pmsimstats biomarker (resting SBP in the prazosin-PTSD application) is continuous and has no defensible cut-point. Dichotomizing at the median or at a clinical threshold loses the within-stratum biomarker variation, attenuates the interaction estimator, and inflates Type II error. In the carryover-sensitivity production data the median split gives a $\hat{L}$ standard error roughly 1.4 to 1.7 times that of the continuous- biomarker linear-mixed coefficient, translating to a 30 to 50 percent power penalty.

2. *Sphericity.* Compound-symmetric within-subject covariance is empirically rejected for time-series symptom data: nearby timepoints are more correlated than distant ones. The AR(1) structure pmsimstats implements explicitly violates sphericity. After Greenhouse-Geisser correction, the test recovers nominal Type I error but loses degrees of freedom and power.

3. *No carryover model.* Strict RM-ANOVA has no provision for residual drug effect during the off-drug period; the model treats off-drug observations as if no drug had been administered. Under the pmsimstats DGP with $t_{1/2} = 1$ week, this misspecification biases $\hat{L}$ toward zero (attenuation by 30 to 35 percent in the production data, comparable to the E1 specification evaluated in the carryover-sensitivity manuscript).

4. *Balanced design required.* Equations (6) and (8) assume equal stratum sizes and equal numbers of within-phase timepoints. Real trials have missing data; classical RM-ANOVA handles missingness only through complete-case deletion. The MMRM framework's likelihood-based handling of missing-at-random (MAR) data[36] is genuinely better-behaved.

5. *Equal time spacing.* Strict RM-ANOVA treats time as a categorical factor with implicitly equal-spaced levels. Real N-of-1 trials have unequally-spaced visits; pmsimstats implements continuous-time AR(1) (`corCAR1`), which the strict-RM-ANOVA framework cannot accommodate.

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These limitations motivate retaining the linear-mixed analysis with continuous-time AR(1) residual correlation as primary, and reporting the strict-RM-ANOVA closed-form F-statistic as a sensitivity comparator. The five violations are not equally consequential: the categorical-biomarker and no-carryover assumptions are the most problematic in the pmsimstats setting, together accounting for a 30 to 50 percent attenuation of the interaction estimator relative to the linear-mixed analysis. The recommendation operationalized in §6 follows this strategy.

4 Cycle-by-period design grid

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The cycle-by-period design grid that varies the number of treatment cycles, on-drug and off-drug period lengths, and on/off symmetry around the[2] hybrid reference is specified in `docs/27-cycle-period-design-sweep-plan.md` and constitutes the second axis of the joint design-by-test sensitivity question framed in §1. The empirical sweep across this grid is deferred to a follow-up paper. We shall note here, in the interest of transparency, that the present paper narrows its scope to the test-procedure axis at the prazosin/PTSD reference design (Hendrickson hybrid open-label plus blinded discontinuation, OL+BDC; $N \in \{25, 35, 70, 100\}$, $\beta_{bm} \in \{0, 0.45\}$), which the simulation infrastructure does support at production rep counts. The §6 recommendation is correspondingly scoped to test-procedure choice rather than the joint (cycle count, period length, test procedure) optimum that the design-by-test sensitivity question requires. The follow-up paper executing the cycle-by-period sweep is the natural extension and is recorded as a portfolio item.

5 Simulation design

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The simulation program follows the ADEMP framework of[1]. The primary estimand is the biomarker-treatment interaction inference under each of the five test procedures. Four operate on the continuous biomarker and share the same interaction estimand $\beta_{bm:D}$ on the symptom scale: a within-subject treatment-effect contrast regressed on the continuous biomarker (the dichotomization-free analogue of the two-sample t-test of equation (11)); the linear-mixed-effects Wald t on $\hat{\beta}_{bm:D}$; the GEE Wald z on $\hat{\beta}_{bm:D}$ with naive sandwich variance; and the GEE Wald z with Mancl-DeRouen small-sample sandwich variance per[34]. The fifth, the strict classical RM-ANOVA F-statistic, operates on the median-dichotomized biomarker and so targets a dichotomized estimand on a different scale. The continuous-biomarker contrast is included specifically to separate the cost of dichotomization (strict RM-ANOVA versus the contrast, both pre-averaged to one value per phase) from the cost of the test procedure proper (the contrast versus the longitudinal linear-mixed and GEE analyses). Primary inferential outputs are power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and Type I error under $\beta_{bm:D} = 0$. Secondary estimands, reported for the four coefficient-bearing arms, are bias of $\widehat{\beta}_{bm:D}$ relative to a high-precision Monte-Carlo reference value of the estimand, the empirical and the mean model-based standard error, and nominal-95% CI coverage. Strict RM-ANOVA is excluded from these coefficient-based measures because its estimand is on the dichotomized scale.

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Performance measures are reported with MCSE alongside every estimate, following Morris et al.'s recommended discipline. Power and Type I error MCSE follow the binomial-proportion form $\sqrt{\widehat{\pi}(1-\widehat{\pi})/n_{\text{reps}}}$. Bias and empirical SE of $\widehat{\beta}_{bm:D}$ are reported with the standard MCSE of the within-replicate mean estimator.

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The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/test-procedure-design-sensitivity/00-ade` which fixes the test-procedure times design-grid factorial across three studies (Study 1: test-procedure comparison at fixed design; Study 2: cycle-by-period grid at fixed test, 240 cells; Study 3: joint design-by-test heatmap, 240 cells) before the production runs commit. The present paper executes a revised Study 1; the executed design departs from the pre-registered Study 1 on several factor levels, and we document every departure in Table 1 below in the interest of full ADEMP transparency.

Table 1: Pre-registered versus executed Study 1. Departures are recorded for transparency; the executed superset of sample sizes (which restores the pre-registered N levels and adds $N = 25$) and the added continuous-biomarker contrast arm respond to referee comments on the first internal draft.

Factor	Pre-registered	Executed (this paper)
Test procedures	3 (RM-ANOVA, LME, GEE-MD)	5 (adds naive GEE and the continuous-biomarker contrast)
Biomarker effect β_{bm}	$\{0, 0.3, 0.6\}$	$\{0, 0.45\}$ (the project-wide reference effect size)
Sample size N	$\{35, 70, 100\}$	$\{25, 35, 70, 100\}$ (superset; adds $N = 25$)
Replicates per cell	1000 (5000 for Type I cells)	1000 power, 2000 Type I

Factor	Pre-registered	Executed (this paper)
Carryover half-life	$t_{1/2} = 3$ days	$t_{1/2} = 0$ (no carryover at the fixed reference)

Units. Carryover half-lives $t_{1/2}$ are quoted in days in this table, following the pre-registration and the main-effect manuscripts of this series. The interaction-focused manuscripts quote $t_{1/2}$ in weeks on the canonical grid $\{0, 0.5, 1.0\}$; the conversion is 1 week $= 7$ days. Timepoints t are weekly in both conventions.

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> In the scope of the present paper (test-procedure axis only at the reference design), the simulation runs on the pmsimstats package data-generating process under the mean-moderation architecture (mean moderation), exercised through `generateData()` and the standard `lme_analysis()` fit, rather than on the reduced `implementations/simple` substrate. The factorial comprises $n_{\text{reps}} = 1000$ per power cell and 2000 per Type I cell, across the 40 cells defined by procedure $\in$ {strict RM-ANOVA, continuous-biomarker contrast, linear-mixed with `corCAR1`, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction} $\times$ $\beta_{bm} \in \{0, 0.45\} \times N \in \{25, 35, 70, 100\}$. Execution uses `parallel::mclapply` with `RNGkind("L'Ecuyer-CMRG")` and `mc.set.seed = TRUE`, so each replicate draws from an independent, reproducible random-number stream. The Mancl-DeRouen correction was implemented inline (no `geesmv`-style external dependency in the production path) by inflating each cluster's residual via $(I - H_{ii})^{-1} r_i$ in the sandwich meat. It produces identical point estimates and larger standard errors than the naive sandwich, and its SE for the interaction coefficient agrees with `geesmv::GEE.var.md` to within approximately 2.5% (mean inline/`geesmv` ratio 1.02, standard deviation 0.02, over 25 probe replicates at $N = 35$, $\beta_{bm} = 0.45$), so the correction magnitude is externally validated rather than merely plausible.

6 Results

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> The simulation program was executed across the 40 cells of the five-procedure-by-effect-by-N factorial in 587 s using 7-core `parallel::mclapply` with `RNGkind("L'Ecuyer-CMRG")` per-stream seeding. Convergence was 100% across all 60{,}000 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; the Morris ADEMP cell-level summary is at `analysis/data/quick-sim/08-test-summary.txt`. With 1000 replicates per power cell the MCSE on a power estimate near 0.5 is approximately 0.016, and with 2000 replicates per Type I cell the MCSE on a rate near 0.05 is approximately 0.005. We begin the results with the test-procedure

comparison at the fixed reference design.

6.1 Test-procedure comparison at fixed design

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We note at the outset that strict RM-ANOVA targets a different estimand from the other four procedures. It operates on a median-dichotomized biomarker, pre-averaged to one value per phase, and so estimates a dichotomized-stratum interaction; the continuous-biomarker contrast, the linear-mixed model, and both GEE variants estimate the interaction coefficient $\beta_{bm:D}$ on the continuous symptom scale. An earlier version of this comparison contrasted strict RM-ANOVA directly with the mixed model and attributed the resulting power gap to the test procedure; that attribution conflated two distinct costs. We therefore introduce the continuous-biomarker contrast of §4, which differs from strict RM-ANOVA only in retaining the continuous biomarker (it shares the pre-averaging and the absence of any AR(1) machinery) and differs from the linear-mixed model only in discarding the within-phase longitudinal structure. The RM-ANOVA → contrast step therefore measures the cost of dichotomization in isolation, and the contrast → linear-mixed step measures the cost of the test procedure proper. Reading the two steps separately is the honest decomposition the earlier direct comparison lacked.

Type I error at the null cell ($\beta_{bm} = 0$), percent ± MCSE, is shown in Table 2; power at $\beta_{bm} = 0.45$ in Table 3.

Table 2: Type I error rate (percent ± MCSE) under $H_0 : \beta_{bm:D} = 0$, 2000 replicates per cell. Nominal level 5%.

Procedure	$N = 25$	$N = 35$	$N = 70$	$N = 100$
Strict RM-ANOVA (dich.)	4.55 ± 0.47	4.15 ± 0.45	5.95 ± 0.53	5.20 ± 0.50
Contrast (continuous bm)	5.25 ± 0.50	4.50 ± 0.46	6.45 ± 0.55	4.60 ± 0.47
Linear-mixed (`corCAR1`)	2.75 ± 0.37	2.05 ± 0.32	3.35 ± 0.40	2.90 ± 0.38
GEE (naive sandwich)	9.70 ± 0.66	8.55 ± 0.63	7.90 ± 0.60	7.40 ± 0.59
GEE (Mancl-DeRouen)	6.35 ± 0.55	5.30 ± 0.50	6.25 ± 0.54	5.90 ± 0.53

Table 3: Power (percent ± MCSE) at $\beta_{bm} = 0.45$, 1000 replicates per cell.

Procedure	$N = 25$	$N = 35$	$N = 70$	$N = 100$
Strict RM-ANOVA (dich.)	35.8 ± 1.52	46.8 ± 1.58	75.3 ± 1.36	91.3 ± 0.89
Contrast (continuous bm)	53.8 ± 1.58	67.7 ± 1.48	91.5 ± 0.88	98.8 ± 0.34

Procedure	$N = 25$	$N = 35$	$N = 70$	$N = 100$
Linear-mixed (`corCAR1`)	56.5 ± 1.57	71.2 ± 1.43	94.5 ± 0.72	99.2 ± 0.28
GEE (naive sandwich)	67.2 ± 1.48	75.9 ± 1.35	94.5 ± 0.72	99.2 ± 0.28
GEE (Mancl-DeRouen)	57.3 ± 1.56	69.1 ± 1.46	92.7 ± 0.82	99.1 ± 0.30

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Four substantive findings emerge from the tables. First, and most consequentially for how the comparison should be read, the power gap between strict RM-ANOVA and the linear-mixed model is predominantly a cost of dichotomizing the biomarker rather than a property of the test procedure. At $N = 25$ the strict RM-ANOVA power is 0.358; the continuous-biomarker contrast, which differs only in retaining the continuous biomarker, reaches 0.538, so dichotomization alone accounts for 0.180 of lost power. The further step to the full linear-mixed model raises power only to 0.565, an increment of 0.027 attributable to the test procedure proper. The decomposition is stable across the grid: the dichotomization component is 0.209 at $N = 35$ $(0.468 \to 0.677)$, 0.162 at $N = 70$ $(0.753 \to 0.915)$, and 0.075 at $N = 100$ $(0.913 \to 0.988)$, while the contrast-to-linear-mixed increment never exceeds 0.04. The earlier framing, which compared strict RM-ANOVA against the mixed model directly and read the difference as a test-procedure effect, therefore overstated the role of the test procedure by roughly a factor of five; the honest statement is that a classical, sphericity-free, AR(1)-free within-subject contrast on the continuous biomarker captures most of the available power, and the mixed model adds a small further increment.

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Second, GEE with the naive sandwich is anti-conservative across the entire sample-size range examined. Its null rejection rate is 0.097 at $N = 25$ and declines only slowly with N to 0.086, 0.079, and 0.074 at $N = 35, 70, 100$ – still roughly $1.5\times$ nominal at $N = 100$. The inflation is therefore not a small-cluster curiosity that disappears at trial-relevant sizes; it persists. The Mancl-DeRouen 2001 small-sample sandwich correction brings the rate within or close to MCSE of nominal at every N $(0.064, 0.053, 0.063, 0.059)$, at a power cost of about ten percentage points at $N = 25$ $(0.672 \to 0.573)$ and seven at $N = 35$ $(0.759 \to 0.691)$. The corrected procedure's apparent power advantage over the linear-mixed model at small N (Table 3) is the residue of its remaining slight over-rejection, not a genuine efficiency gain.

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Third, the secondary estimands (Table 4) locate the procedures' behavior in the standard-error calibration that the rejection rates only summarize. At $\beta_{bm} = 0.45$, the continuous-biomarker contrast and the Mancl-DeRouen GEE are well-calibrated: their nominal-95% CI coverage is 93–95% and their mean model-based SE is within a few percent of the empirical SE. Naive GEE under-covers at small N (90.0% at $N = 25$, 91.5% at $N = 35$) because its model-based SE is 9 to 12% smaller than the empirical SE, the calibration counterpart of its Type I inflation. The linear-mixed model over-covers (97.0 to 97.7%) because its model-based SE exceeds the empirical SE by 11 to 16%; this is the same conservatism visible in its sub-nominal Type I error.

Table 4: Secondary estimands at $\beta_{bm} = 0.45$ for the four coefficient-bearing arms (the interaction estimand is $\beta_{bm:D} = -2.21$, the Monte-Carlo reference value). Coverage is of the nominal-95% Wald interval; SE ratio is mean model-based SE divided by empirical SE.

Arm	Coverage $N = 25$	Coverage $N = 100$	SE ratio $N = 25$	SE ratio $N = 100$	Max |rel. bias|
Contrast (continuous bm)	94.9	94.8	0.99	0.98	3.0%
Linear-mixed (`corCAR1`)	97.7	97.0	1.16	1.11	2.6%
GEE (naive sandwich)	90.0	93.3	0.91	0.98	3.8%
GEE (Mancl-DeRouen)	94.3	94.0	1.07	1.02	3.8%

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Fourth, the comparison is not contaminated by point-estimate bias. The relative bias of $\hat{\beta}_{bm:D}$ is below 4% of the estimand $|\beta_{bm:D}| = 2.21$ for the contrast, the linear-mixed model, and both GEE variants in every cell, so the differences in power and coverage reflect variance estimation and information use rather than systematic mis-estimation. The linear-mixed procedure's Type I error sits below nominal throughout (0.021 to 0.034), the conservatism already visible in its over-coverage and its model/empirical SE ratio above one; it is consistent with `corCAR1` degrees-of-freedom accounting under the short ($T = 8$) serial series. A Kenward-Roger[38] or Satterthwaite[39] small-sample degrees-of-freedom correction would address the conservatism and recover the small power increment it forgoes; this extension is recorded as a follow-up item.

7 Design-and-test recommendation

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The recommendation table below maps test procedure choice to typical chronic-condition N-of-1 trial scenarios at the prazosin/PTSD reference design (Hendrickson hybrid OL+BDC, biomarker-treatment interaction estimand). The cycle count and period length are held at the reference values defined in §3 because the joint optimization over those axes is deferred to the follow-up paper. We present the recommendation as a table of preferred (procedure, scenario) combinations rather than as a single 'best' procedure, because the preferred choice depends on features of the application that the analyst must determine. The overriding recommendation, however, is design-agnostic: retain the biomarker as a continuous predictor, since dichotomization forfeits the large majority of the available power (§5, Table 3).

Scenario	Preferred procedure	Rationale
Default (continuous biomarker, any N)	linear-mixed (`corCAR1`)	Highest power within nominal Type I; `corCAR1` matches AR(1) physiology; well-behaved across N
Lightweight / transparent classical analysis	continuous-biomarker contrast	Within 0.03 of the mixed model on power, nominal Type I, no covariance modeling required
$N \leq 35$, concern about LME conservatism	linear-mixed (`corCAR1`) with Kenward-Roger	LME sub-nominal and over-covers; KR addresses this without inflating Type I
Robust-SE-required setting (e.g. unmodeled clustering)	GEE Mancl-DeRouen	Restores nominal Type I and 93–95% coverage; modest power cost
Reviewer-mandated 'classical' analysis	strict RM-ANOVA	Dichotomized sensitivity comparator; report alongside a continuous-biomarker analysis with the power-loss decomposition
Naive GEE (no small-sample correction)	not recommended	Type I error 1.5–1.9× nominal across all N; under-covers

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The dominant message is that the biomarker should be analyzed as a continuous predictor: dichotomizing it for a strict RM-ANOVA forfeits the large majority of the power gap that the present study isolates, while modeling choices among the continuous-biomarker procedures move power only at the margin. Within that class, the linear-mixed analysis with continuous-time AR(1) residual correlation is the preferred default for the biomarker-treatment interaction in aggregated N-of-1 trials at the parameter scale evaluated here, with the dichotomization-free within-subject contrast a close and transparent alternative that requires no covariance modeling. Strict RM-ANOVA remains useful as a dichotomized sensitivity comparator, as the §2 closed-form derivation makes explicit, but it should be reported alongside a continuous-biomarker analysis and the power-loss decomposition rather than on its own. Naive GEE should not be used at the sample sizes examined without a small-sample correction; the Mancl-DeRouen correction restores nominal Type I error and 93–95% coverage at a moderate power cost.

8 Discussion

8.1 What the Monte Carlo study settles

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We begin the discussion by summarizing what the Monte Carlo study settles at the resolution it was run. The run on the 40-cell five-procedure factorial (procedure $\in$ {strict RM-ANOVA, continuous-biomarker contrast, linear-mixed with `corCAR1`, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction} $\times$ $\beta_{bm} \in \{0, 0.45\} \times N \in \{25, 35, 70, 100\}$, with 1000 replicates per power cell and 2000 per Type I cell) was executed in 587 s with 100% convergence across all 60{,}000 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; Morris ADEMP summary at `08-test-summary.txt`. With 1000 power replicates the MCSE on a power estimate near 0.5 is approximately 0.016 and with 2000 Type I replicates the MCSE on a rate near 0.05 is approximately 0.005. The Mancl-DeRouen 2001[34] correction was implemented inline by inflating each cluster's residual via $(I - H_{ii})^{-1} r_i$ in the sandwich meat; it produces identical point estimates and larger standard errors than the naive sandwich, and its SE for the interaction coefficient agrees with `geesmv::GEE.var.md` to within approximately 2.5% (mean inline/`geesmv` ratio 1.02 over 25 probe replicates), so the correction magnitude is externally validated.

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The study resolves four substantive questions at this resolution. First, and correcting the emphasis of an earlier draft, the power gap between strict RM-ANOVA and the linear-mixed model is predominantly the cost of dichotomizing the biomarker, not a property of the test procedure. The continuous-biomarker contrast, which differs from strict RM-ANOVA only in retaining the continuous biomarker, recovers 0.180 of the 0.207 gap at $N = 25$ ($0.358 \to 0.538$ of $0.358 \to 0.565$); the further step to the full mixed model contributes only 0.027. The same decomposition holds across the grid (dichotomization 0.08 to 0.21, test procedure at most 0.04). The earlier direct comparison of RM-ANOVA against the mixed model therefore overstated the test-procedure effect by roughly fivefold; the corrected statement is that keeping the biomarker continuous, not adopting the mixed model per se, is what recovers the power.

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Second, naive GEE is anti-conservative across the entire sample-size range, not merely at the smallest N. Its null rejection rate is 0.097 at $N = 25$ and falls only to 0.086, 0.079, and 0.074 at $N = 35, 70, 100$, remaining roughly 1.5× nominal even at $N = 100$. The Mancl-DeRouen correction returns the rate to within or near MCSE of nominal at every N $(0.064, 0.053, 0.063, 0.059)$, at a power cost of about ten percentage points at $N = 25$ $(0.672 \to 0.573)$ and seven at $N = 35$ $(0.759 \to 0.691)$. Naive GEE's nominally higher power than the linear-mixed model at small N is an artifact of its residual over-rejection rather than a real gain, and disappears once the size distortion is corrected.

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Third, the standard-error calibration of the four coefficient-bearing arms is what underlies the rejection-rate differences. The continuous-biomarker contrast and the Mancl-DeRouen GEE are well-calibrated, with 93 to 95% nominal-95% coverage and model-based standard errors within a few percent of the empirical standard error. Naive GEE under-covers at small N (90.0% at $N = 25$) because its model-based SE is 9 to 12% too small, the same defect that drives its Type I inflation. The linear-mixed model over-covers (97.0 to 97.7%) because its model-based SE exceeds the empirical SE by 11 to 16%, the calibration counterpart of its sub-nominal Type I error. All four arms are approximately unbiased (relative bias below 4%), so these are pure variance- estimation effects.

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Fourth, the linear-mixed procedure's Type I error sits below nominal throughout (0.021 to 0.034), conservative rather than nominal and consistent with `corCAR1` degrees-of-freedom accounting under the short ($T = 8$) serial series. The conservatism is a variance-calibration effect, visible in the over-coverage and the model/empirical SE ratio above one of Table 4, not a point-estimate effect. A Kenward-Roger[38] or Satterthwaite[39] small-sample degrees-of-freedom correction would address it and recover the small power increment it forgoes, and we record this as a follow-up item; the recommendation table in §6 already lists the Kenward-Roger variant for small-N use.

9 Conclusions

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In conclusion, three points are to be emphasized. First, the analyst's most consequential choice for the biomarker-treatment interaction in aggregated N-of-1 trials is to keep the biomarker continuous rather than to dichotomize it for a classical RM-ANOVA. The continuous-biomarker contrast recovers about four-fifths of the apparent RM-ANOVA-to-mixed-model power gap (dichotomization costs 0.08 to 0.21 of power across the sample sizes evaluated, while the test procedure proper adds at most 0.04); the linear-mixed analysis with continuous-time AR(1) residual correlation has the highest power within nominal Type I among the continuous-biomarker procedures, with the dichotomization-free within-subject contrast a close alternative. Naive GEE with the standard sandwich estimator inflates Type I error to between 1.5× and 1.9× nominal across the whole $N \in \{25, 35, 70, 100\}$ range, disqualifying its rejection rate as a power summary; the Mancl-DeRouen 2001 small-sample sandwich correction restores nominal Type I control and 93–95% coverage at a moderate (about 10-percentage-point) power cost, making it the operational GEE comparator rather than the naive sandwich.

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Second, the closed-form strict-RM-ANOVA test of the biomarker-treatment interaction (§2) reduces to a two-sample t-test on the within-subject treatment differences under sphericity and balance. The reduction makes the strict-RM-ANOVA result analytically transparent and connects the simulation findings to the classical split-plot literature. It also pins down what does and does not limit the test here. The strict RM-ANOVA arm we evaluate pre-averages each phase to a single value, so its within-subject covariance is 2×2 and sphericity holds by construction; consistent with this, its Type I error is at nominal (Table 2), and its power deficit is wholly attributable to the median dichotomization of the biomarker, as the continuous-biomarker contrast demonstrates. The sphericity penalty that motivates much of §2 applies instead to the alternative classical analysis that treats the full $T = 8$ series as repeated within-subject levels: there the AR(1) residual correlation violates sphericity and a Greenhouse-Geisser correction recovers nominal Type I only at a degrees-of-freedom cost, which the linear-mixed analysis with `corCAR1` avoids by modeling the AR(1) structure directly. Both considerations point the same way – away from a dichotomized classical analysis – but through different mechanisms, and we are careful to keep them distinct.

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Third, the cycle-by-period design grid that constitutes the second axis of the joint design-by-test sensitivity question framed in §1 is deferred to a follow-up paper. The infrastructure for the sweep is in place at `analysis/scripts/test-procedure-design-sensitivity/`, and the design specification is committed at `docs/27-cycle-period-design-sweep-plan.md`. The follow-up paper will deliver the joint (cycle count, period length, test procedure) recommendation table that the present paper's §6 narrows to the test-procedure-only axis.

10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

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The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths relevant to this manuscript are listed in the Reproducibility section below.

13 Reproducibility

This manuscript's simulation was executed with R 4.6.x; the project also ships a zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`) for container-based reproduction. Principal packages used by the Study 1 pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `geepack` (GEE fitting) plus an inline Mancl-DeRouen 2001 small-sample sandwich correction, `geesmv` (independent benchmark of that correction), `data.table`, and `parallel` (`mclapply` for the parameter-grid sweep), with `rmarkdown` plus `bookdown` for the manuscript build. The Study 1 simulation runs on the pmsimstats package data-generating process under the mean-moderation architecture via `generateData()` and `lme_analysis()`, not on the reduced `implementations/simple` substrate.

Random-number generator. The driver sets `RNGkind("L'Ecuyer-CMRG")` and `set.seed(20260507)`, then runs the cell-by-replicate jobs through `parallel::mclapply` with `mc.set.seed = TRUE`, so each replicate draws from an independent, reproducible random-number stream.

Data and code availability. Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/08-test-replicates.rds`; the extended Morris ADEMP cell-level summary (rejection rate, MCSE, convergence, bias, empirical and model SE, coverage) at the companion `08-test-summary.txt`, with the Monte-Carlo reference estimand at `08-test-betatrue.txt` and the Mancl-DeRouen benchmark at `08-test-mancl-benchmark.txt`. The driver is `analysis/scripts/test-procedure-design-sensitivity/01` (superseding the earlier `analysis/scripts/quick-sim/08-test-procedure-prototype.R`). The pre-registered ADEMP plan, and the deviations recorded in Table 1, are at `analysis/scripts/test-procedure-design-sensitivity/00-ademp-pre-registration.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/08-test-procedure-design-sensitivity/`.

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